OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2024–2025

Exam Codes	4BA1, 3BA1, 4BME1, 4BPS1, 1OA2, 1OA3, 4BCT1, 4BMS2, 4BS2
Exam(s)	3rd and 4th year (Arts and Science), Study Abroad / International Exchange
Module Codes	CS402
Module	**Cryptography**
Paper No.	1
External Examiner	Prof. Colva Roney-Dougal
Internal Examiners	Dr Niall Madden Dr Koushik Paul*
Instructions:	**Answer all four questions.** **Each question carries 25 marks.** **All workings must be shown.**
Duration	2 Hours
Number of Pages	3 (including this page)
Discipline	Mathematics
Requirements:	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No
Graph paper	No

Q1. (a) A treasure hunt challenge uses a shift cipher over a 27-letter alphabet:

$$\mathtt{A} = 0,\ \mathtt{B} = 1,\ \ldots,\ \mathtt{Z} = 25,\ \llcorner\!\lrcorner = 26.$$

A group of adventurers found a coded message:

`EULOOLDQWCZRUN`

The game master revealed that in a previous puzzle, the word `TREASURE` was encoded as `WUHDVXUH` using the same shift cipher. Using this information, determine the decoded message.

[5 marks]

(b) A *Hill cipher* over $\mathbb{Z}_{27}$ (where $\mathtt{A} = \mathtt{0}, \mathtt{B} = \mathtt{1}, ..., \mathtt{Z} = \mathtt{25}$, and $\llcorner\!\lrcorner = \mathtt{26}$) was used to encrypt a message. The ciphertext obtained is: `LPDG`. The encryption used the key matrix:

$$K = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$$

Find the corresponding plaintext.

[10 marks]

(c) State *Kerckhoff's principle.*

[5 marks]

(d) A digital lock system encrypts passcodes using a (1-dimensional) *affine cipher* over $\mathbb{Z}_{900}$ (passcodes range from 0 to 899). Find the total number of possible encryption keys for this system.

[5 marks]

Q2. (a) State (without proof) *Shannon's Theorem* on perfectly secure cryptosystems.

[5 marks]

(b) Consider a symmetric cryptosystem with non-empty finite sets $\mathcal{P}, \mathcal{C}$, and $\mathcal{K}$ of plaintexts, ciphertexts, and keys, respectively and with encryption functions $e_k : \mathcal{P} \to \mathcal{C}$ and decryption functions $d_k : \mathcal{C} \to \mathcal{P}$ indexed by keys $k \in \mathcal{K}$. Suppose that our cryptosystem is perfectly secure. Show that $|\mathcal{K}| \geq |\mathcal{C}| \geq |\mathcal{P}|$.

[10 marks]

(c) Briefly describe *block ciphers* and *stream ciphers* with examples.

[5 marks]

(d) Explain what is meant by a *linear feedback shift register* of length L.

[5 marks]

Q3. (a) Describe the key generation, encryption, and decryption steps of the *RSA cryptosystem.*

[5 marks]

(b) Determine with proper justification whether the following pairs of integers are valid RSA keys:

(i) $(n, e) = (299, 17)$

(ii) $(n, e) = (323, 9)$

(iii) $(n, e) = (385, 55)$

[6 marks]

(c) Alice encrypts the message $m = 23$ using the RSA cryptosystem and Bob's public key $(n, e) = (77, 17)$. Find the corresponding ciphertext (as an element of $\mathbb{Z}_{77}$).

[5 marks]

(d) What does it mean for a number n to be a *pseudoprime*? Give the definition of a *Carmichael number*.

[4 marks]

Q4. (a) Describe Fermat's primality test (which is passed by every prime).

[5 marks]

(b) Describe the method of *fast exponentiation* for computing $a^e \bmod n$, where $a, e, n \in \mathbb{Z}$ with $e, n > 1$.

[5 marks]

(c) Alice and Bob use the cyclic group $\mathbb{Z}_{61}^{\times}$ (multiplicative group modulo 61) with generator $g = 2$ to perform a *Diffie-Hellman* key exchange. Alice secretly chooses $a = 5$ and Bob secretly chooses $b = 7$. Determine their shared key (as an element of $\mathbb{Z}_{61}^{\times}$).

[8 marks]

(d) Describe the key generation, encryption, and decryption steps of the *ElGamal cryptosystem.*

[6 marks]

(e) Consider the elliptic curve E over $\mathbb{Z}_7$ defined by the equation:

$$y^2 = x^3 + 2x + 3,$$

with the point at infinity $\mathcal{O}$.

(i) Determine the set of points $E(\mathbb{Z}_7)$ on this elliptic curve, including the point at infinity $\mathcal{O}$.

(ii) Find all points of order 2 in $E(\mathbb{Z}_7)$.

[6 marks]